

Shuyu Wang

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<https://github.com/shuyuw-astro>

Research Interests	Milky Way dynamics, Magellanic Clouds, galactic dynamics, galactic archaeology		
Education	Ph.D. in Astronomy	Sep. 8 2023 - Present	
	Leiden Observatory, Leiden University, Leiden, Netherlands		
	Master of Science in Physical Sciences	Sep. 2020 - Dec. 2021	
	University of Chicago, Chicago, Illinois, USA		
	Bachelor of Science in Physics	Aug. 2016 - Jun. 2020	
	Southern University of Science and Technology, Shenzhen, China		
	Undergraduate Exchange Program	Aug. 2019 - Dec. 2019	
	University of California Berkeley, Berkeley, California, USA		
Awards and Honors	Student Scholarship at the University of Chicago (\$18,472)	2020	
	Outstanding Undergraduate Thesis	2020	
Publications	Wang S., Brown A.G.A., Debattista V.P., Khachaturyants T. 2025, <i>Phase Spirals Induced by the Gas Warp</i> , A&A, 710, A120 (2026).		
	Wang S., Necib L., Ji A.P., et al. 2023, <i>High-Resolution Chemical Abundances of the Nyx stream</i> , ApJ 955 129.		
	Wang S. 2020, <i>Analysis of Nonlocal Gravity in Terms of Auxiliary Fields</i> (undergraduate thesis).		
Research Experience	Calculating the Tilting Rate of the Milky Way Disc	Sep. 2023 - Present	
	Leiden University		
	Mentored by Dr. A. Brown		
	• Study the kinematics of the Milky Way disc stars using N-body simulation of isolated galaxy • Applying the Fourier decomposition method to the vertical velocity field of the simulated star particles to analyze the various kinematic components (stellar warp, spiral arms, bending waves, tilting) • Using py-ananke to generate mock Gaia catalog, and study the effect of observational errors and selection function		

- Using the Gaia DR3 6-dimensional phase space information of the Milky Way disc stars to estimate the tilting rate of the disc, and compare it with results obtained from the simulation

PhD Preparation

Oct. 2022 - Aug. 2023

- Ph.D. application
- Revised manuscript on the chemical abundances of the Nyx stream based on the referee report. The updated article is now published by the ApJ

Sensitivity of CSST Telescope for Exoplanet Microlensing Event Detection Feb. 2022 - Jun. 2022

Tsinghua University

Mentored by Dr. W. Zhu

- Calculated the probability of detecting caustic crossing events with CSST by simulating such events for varying parameters (e.g., mass ratio and separation) and time sequences using MulensModel
- Probed the sensitivity of CSST telescope for detecting exoplanet microlensing events by calculating the minimum mass of a detected exoplanet in a simulated event

High-resolution chemical abundances of the Nyx stream

Sep. 2021 - Oct. 2022

University of Chicago

Mentored by Dr. A. P. Ji, Dr. L. Necib

- Performed high-resolution spectroscopic studies using SMH on 34 stars in the Nyx stream observed with Magellan/MIKE and Keck/HIRES to analyze their chemical abundances. Compared the abundances with those of Galactic thick disk stars to determine the origin of Nyx and its possible implication on the local dark matter phase-space distribution under the guidance of Dr. A. P. Ji
- Measured absolute chemical abundances of the Nyx stars using a combination of the equivalent widths and spectral synthesis methods
- Measured final chemical abundances of the Nyx stream using a differential abundance analysis method
- Compared the results with the Galactic thick disk stars from Bensby et al. 2014 to determine the origin of Nyx and its implication on the local dark matter phase-space distribution
- Integrated the abundance results with kinematics from Gaia DR2 and Gaia eDR3 under the guidance of Dr. L. Necib. Wrote and submitted the manuscript "High-Resolution Chemical Abundances of the Nyx Stream" to *The Astrophysical Journal* under the guidance of Dr. L. Necib and Dr. A. P. Ji

Deep Learning Basics and Strong Lensing

Sep. 2021 - Dec. 2021

University of Chicago

Mentored by Dr. B. Nord

- Studied basic concepts of deep learning with Keras and TensorFlow
- Familiarized with the deep-learning-based strong lensing finder CMU DeepLens
- Wrote a script to output the loss function plot, confusion matrix and roc curve of test deep learning models

Prediction of Crime Rates and Primary Crime Type in Chicago Using Machine Learning Sep. 2021 - Dec. 2021

University of Chicago

Mentored by Dr. R. Willett

- Obtained a dataset of reported incidents of crime in Chicago from 2016 to 2020 and divided the data into a training set and a validation set using the Cross Validation (CV) technique, labelling the associated crime types with numbers from 1 to 17
- Reduced noise and predicted the change in crime rates in Chicago using polynomial fitting
- Used the method of Principal Component Analysis (PCA) to reduce the dimension of the crime type data matrix to two
- Used the Least Squares (LS) method to derive the matrix of predicted crime type and compared it with the test matrix to verify the accuracy of the prediction
- Wrote a final report on the results

Pulse Profile of Pulsar J1518+4904

Apr. 2021

Key Laboratory of Five-hundred-meter Aperture Spherical radio Telescope (FAST)

Mentored by Dr. J. Lu

- Analyzed FAST observational data of pulsar J1518+4904
- Calculated the frequency of each data point prior to the Doppler effect correction and the corresponding dispersion delay time
- Employed the linear interpolation method to produce the pulse profile of pulsar J1518+4904 in one period, generated a Frequency-Phase diagram of the pulse and reduced the noise in the background manually
- Gave a presentation on the results

Galactic and Extragalactic Magnetic Fields

Sep. 2020 - Jan. 2021

University of Chicago

Mentored by Dr. H.-W. Chen

- Studied empirical methods for determining galactic and extragalactic magnetic fields
- Analyzed each method with observational data from the Milky Way and extragalactic radio sources
- Wrote a final report and gave a presentation on the results

Accretion Disk Models around Active Galactic Nuclei

Sep. 2019 - Dec. 2019

University of California Berkeley

Mentored by Dr. J. Lu

- Derived the equation of state and energy transfer mechanism of accretion disk models around active galactic nuclei
- Analyzed the emission line of AGN accretion disk wind
- Extracted data from quasi-stellar objects and compared their optical spectra with extant stellar spectra
- Wrote a final report and presented the results

Nonlocal Gravity

Aug. 2019 - Jun. 2020

Southern University of Science and Technology

Mentored by Dr. L. Modesto

- Studied the theory of nonlocal gravity as an extension of local higher derivative gravity which preserves both renormalizability and unitarity
- Substituted higher-derivative operators with infinite-derivative operators (“non-local form factors”) and introduced appropriate auxiliary fields to the action of nonlocal gravity
- Analyzed future prospects of such a theory, specifically, the possibility of singularity avoidance

	• Wrote an undergraduate thesis titled “Analysis of Nonlocal Gravity in Terms of Auxiliary Fields”	
Skills	• Programming: Python, TensorFlow (Basics), JAX, SQL • Operating system: Mac OS, Linux • Software: LaTeX, Git • Observing: – 3 remote observing nights on the Magellan (Clay) Telescope using the MIKE spectrograph, mentored by Dr. A. P. Ji (primary observer) • Language: Chinese (Native), English	Oct. 2021
Posters	<i>Tilting rate of the Milky Way Disk</i> • July 2024: European Astronomical Society Annual Meeting (EAS), Padova Congress	
Academic Visit	• Jan & Dec 2024: University of Lancashire (hosted by Victor P. Debattista)	
Conferences and Workshop	• <i>EDUCADO-MWGaiaDN training school on astro-AI and ML</i>, Ghent Mar. 2026 • <i>ESR School on transferable skills: managing, entrepreneurial and IPR</i>, University of Barcelona Jun. 2025 • <i>Galaxy modelling + galactic centre workshop</i>, Guildford University Dec. 2024 • <i>ESR school: Introduction to Astro-statistics and data visualisation</i>, University of Coimbra Sep. 2024 • <i>European Astronomical Society Annual Meeting</i>, Padova Congress – Poster and short presentation “Tilting Rate of the Milky Way Disc” Jul. 2024 • <i>The Art of Astrometry and Computation</i>, Lund University Jun. 2024 • <i>MWGaia Induction School</i>, Leiden – Project Presentation “Vertical Velocities in the Mock Gaia Catalog” Jan. 2024	
Teaching Experience	<i>Teaching assistant</i> , Leiden University • BSc course ‘Astronomical Observing techniques’ (three times)	2024 - 2026
Leadership Experience	<i>Organizing Committee</i> , EDUCADO-MWGaiaDN training school on astro-AI and ML Mar 2026	